Historical strong partial packet: state-norming units

Run `fa_banach_001`, lane 5

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Historical Status

This note records the intermediate strong partial result obtained after the unital $C(K)$-subspace packet and before the final all-nonzero-E qualitative theorem. It is superseded by the parent full packet, but remains the same-modulus refinement.

Theorem

Theorem 1. *Let E be a Banach space and suppose that there is $e \in S_E$ such that*

$$K_e = \{e^* \in B_{E^*} : e^*(e) = 1\}$$

norms E. Let X and Y be Banach spaces. If $(X, E\widehat{\otimes}_\varepsilon Y)$ has the BPBp with function η, then (X, Y) has the BPBp with the same function η.

Proof

The set K_e is weak-star compact, and the map

$$\Gamma : E \to C(K_e), \qquad \Gamma(a)(e^*) = e^*(a),$$

is an isometry because K_e norms E. Moreover $\Gamma(e) = 1_{K_e}$. Thus, by injectivity of the injective tensor norm,

$$E\widehat{\otimes}_\varepsilon Y \hookrightarrow C(K_e)\widehat{\otimes}_\varepsilon Y \subset C(K_e, Y)$$

isometrically.

Fix $\varepsilon > 0$. Given $T \in S_{\mathcal{L}(X,Y)}$ and $x_0 \in S_X$ with $\|Tx_0\| > 1 - \eta(\varepsilon)$, define $J(y) = e \otimes y$ and $\widetilde{T} = J \circ T$. The map $J : Y \to E\widehat{\otimes}_\varepsilon Y$ is an isometry. Applying BPBp for $(X, E\widehat{\otimes}_\varepsilon Y)$, choose $\widetilde{S}$ and $x_1 \in S_X$ with

$$\|\widetilde{S}\| = \|\widetilde{S}x_1\| = 1, \qquad \|\widetilde{S} - \widetilde{T}\| < \varepsilon, \qquad \|x_1 - x_0\| < \varepsilon.$$

The continuous Y-valued function $\widetilde{S}x_1$ on compact K_e attains its norm at some $e_1^* \in K_e$. Let

$$S = (e_1^* \otimes \mathrm{Id}_Y) \circ \widetilde{S}.$$

Then $\|S\| \le 1$, and

$$\|Sx_1\| = \|(\widetilde{S}x_1)(e_1^*)\| = 1.$$

Since $e_1^*(e) = 1$, evaluation fixes $J(Y)$, so

$$\|S - T\| \le \|\widetilde{S} - \widetilde{T}\| < \varepsilon.$$

Hence (X, Y) has the BPBp with the same modulus.

Supersession Note

The final packet proves qualitative BPBp descent for every nonzero E. The argument there evaluates over the whole dual ball B_{E^*} and rotates a scalar whose modulus is close to one, causing a factor-two modulus loss. The state-norming-unit hypothesis removes that loss.